PS3 — Financial Distress Prevention

Existing Features + Features We Can Add + Possible Advanced Features

Project direction: Financial Reality Engine + Decision Twin — detect distress early, understand the cause, simulate possible futures, and recommend the safest intervention before a financial crisis.

A. EXISTING / ALREADY COMMON FEATURES

1. Early Warning System (EWS)

- Monitor repayment delays, overdue amounts, missed EMIs and delinquency signals.
- Track changes in balances, cash flow, utilisation and account behaviour.
- Generate risk alerts when financial stress indicators cross thresholds.

2. AI / ML Credit Risk Prediction

- Probability-of-default or delinquency prediction.
- Risk scoring from transaction, repayment, credit and financial-history data.
- Classification of customers into low / medium / high-risk groups.

3. Real-Time Financial Monitoring

- Continuous monitoring of customer accounts and financial behaviour.
- Cash-flow trend monitoring.
- Alerts for unusual or deteriorating financial patterns.

4. Account Aggregator (AA)-Based Data Access

- Consent-based access to financial information from participating institutions.
- Potential to combine information across accounts for cash-flow and credit monitoring.
- Useful foundation for continuous financial-health monitoring.

5. Automated Alerts & Notifications

- Payment reminders, overdue alerts and risk notifications.
- Proactive customer communication when warning signals appear.

6. Collections / Recovery Workflows

- Prioritise customers for collection.
- Trigger collection actions based on risk or delinquency.
- Track collection status and outcomes.

7. Debt / Financial Health Dashboard

- Display loans, EMIs, balances, spending, income and financial-health indicators.
- Summarise current financial position for customers or bank staff.

8. Restructuring / Payment Relief Workflows

- Support processes such as payment rescheduling, restructuring or other corrective actions.
- Route distressed accounts to appropriate bank workflows.

9. AI Financial Guidance / Coaching

- Personalised financial tips and recommendations.
- Budgeting, spending and debt-management guidance.

B. FEATURES ALREADY IN OUR PROJECT CONCEPT

10. Financial Reality Engine

- Create one unified financial state from income, expenses, loans, EMIs, receivables, payables and recurring obligations.
- Move beyond a single credit score toward a broader view of financial reality.

11. Distress Detection + Why Engine

- Detect early distress and explain the main contributing factors.
- Example causes: falling inflows, rising debt burden, delayed receivables, low cash buffer or upcoming obligation pressure.

12. Decision Twin

- Create a simulated financial state and test alternative actions.
- Compare how different interventions affect liquidity, debt burden and distress risk.

13. Intervention Recommendation

- Recommend a suitable action instead of automatically pushing additional credit.
- Possible actions include collecting receivables, reducing expenses, rescheduling payments or reviewing debt.

14. Confidence + Explainability Layer

- Show risk level, confidence, contributing factors and supporting evidence.
- Flag low-confidence cases for human review.

15. Human-in-the-Loop Review

- Bank officer or financial advisor reviews significant recommendations before action.
- Keep automated decisions auditable and controllable.

16. Consent + Audit Layer

- Consent-first data access.
- Maintain evidence and audit trails for important predictions and recommendations.

C. NEW FEATURES THAT COULD DIFFERENTIATE OUR PROJECT

17. Obligation Collision Radar ■

- Map expected cash inflows against future obligations on a timeline.
- Identify the exact period when expected available cash may be insufficient.
- Show which inflow and obligation create the collision.

18. Financial Survival / Critical Date ■

- Estimate a customer's cash runway or projected critical liquidity date.
- Answer: 'How long can the customer safely meet essential obligations?'

19. Distress Root-Cause Tree ■

- Break one risk alert into causes such as cash-flow stress, debt stress, expense stress and liquidity gaps.
- Show evidence behind each cause.

20. What-If Financial Simulator ■

- Test scenarios such as EMI rescheduling, expense reduction, receivable collection, delayed payments or new borrowing.
- Compare projected outcomes rather than giving a single static recommendation.

21. Least-Harm Intervention Optimizer ■

- Rank interventions by their ability to prevent distress while minimising additional financial burden.
- Allow the system to recommend 'no new loan' when additional borrowing worsens the situation.

22. Intervention Sequencer ■

- Turn recommendations into an ordered action plan: what to do today, this week and before the critical date.
- Recalculate after each action.

23. Multi-Lender Debt View

- Show total known debt obligations across consented financial relationships.
- Estimate combined EMI/debt-service pressure instead of viewing one lender in isolation.

24. Receivable Collectability Prediction

- Distinguish total receivables from realistically collectible near-term cash.
- Use invoice age and historical payment behaviour to estimate expected collection timing.

25. Financial Stress Testing

- Simulate revenue drops, delayed customer payments, increased expenses or multiple obligations becoming due together.
- Identify the customer's vulnerability under different shocks.

26. 'Should We Give Another Loan?' Guardrail

- Check whether new credit solves the root cause or merely increases debt pressure.
- If borrowing is harmful, recommend alternative interventions.

D. ADVANCED / FUTURE FEATURES

27. Outcome Learning Loop

- Track what happened after an intervention.
- Learn which interventions work for which distress patterns.
- Use actual outcomes to improve future recommendations.

28. Dynamic Risk Timeline

- Show how the customer moved from healthy → vulnerable → stressed → critical.
- Make deterioration and intervention timing visually clear.

29. Adaptive Intervention Re-Ranking

- Recalculate recommendations whenever new transactions, payments or data arrive.
- Change the action plan as the financial state changes.

30. Financial Dependency / Concentration Risk

- Detect dependence on a small number of customers, income sources or lenders.
- Highlight risks that may not appear in a simple cash-flow score.

31. Early Payment Probability

- Predict the likelihood and timing of incoming payments rather than assuming every receivable arrives on time.
- Use this forecast inside the Decision Twin.

32. Responsible AI / Fairness Monitor

- Monitor model performance, false positives and false negatives across relevant customer groups.

- Maintain governance metrics and review triggers.

33. Data Completeness & Reliability Score

- Show whether the financial picture is based on complete, recent and consistent data.
- Lower confidence when important accounts or obligations are missing.

34. Intervention Evidence Card

- For every recommendation, show: reason, evidence, expected benefit, downside, confidence and assumptions.
- Makes the recommendation easier for bank staff to audit.

35. Crisis Escalation Workflow

- Automatically route severe cases to specialised human support.
- Differentiate early guidance from cases requiring urgent restructuring or financial counselling.

E. FEATURES TO PRIORITISE FOR THE HACKATHON MVP

Core demo — MUST HAVE

- Financial Reality Engine
- Early Distress Detection
- Why / Root-Cause Engine
- Obligation Collision Radar
- Decision Twin / What-If Simulator
- Least-Harm Intervention Recommendation

Strong supporting features — SHOULD HAVE

- Critical liquidity date / cash runway
- Multi-lender debt view
- Receivable collectability
- Stress testing
- Confidence score
- Human review
- Consent + audit trail

Future roadmap — CAN ADD LATER

- Outcome learning loop
- Adaptive intervention sequencing
- Fairness monitor
- Advanced concentration risk
- Production AA integrations
- Continuous post-intervention monitoring

F. THE CORE DIFFERENTIATION

What we should NOT claim as the main innovation

- AI predicts default.
- AI detects risky customers.
- AI sends early warnings.

- AI analyses bank transactions.

What we should claim as the stronger innovation

- **Financial Reality → Distress Diagnosis → Obligation Collision → Decision Twin → Least-Harm Intervention → Crisis Prevention.**
- The key question changes from 'Who is risky?' to 'What is going wrong, when will it become critical, what happens under each possible intervention, and what is the safest action?'

Note: 'Existing / already common' means the capability is present in the broader banking, lending, EWS, AA or financial-coaching landscape. 'New / possible' means a proposed feature for our project; it should be validated with a prototype and competitor review before being described as uniquely novel.